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DIGITAL EXHIBITS

FocusFlow

The Executive Function OS

An adaptive digital platform that reimagines how neurodivergent users plan, prioritize, and initiate tasks by meeting the brain where it actually is.

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FocusFlow Dashboard

The primary interface, showing energy check-in, Next Best Action, and task queue.

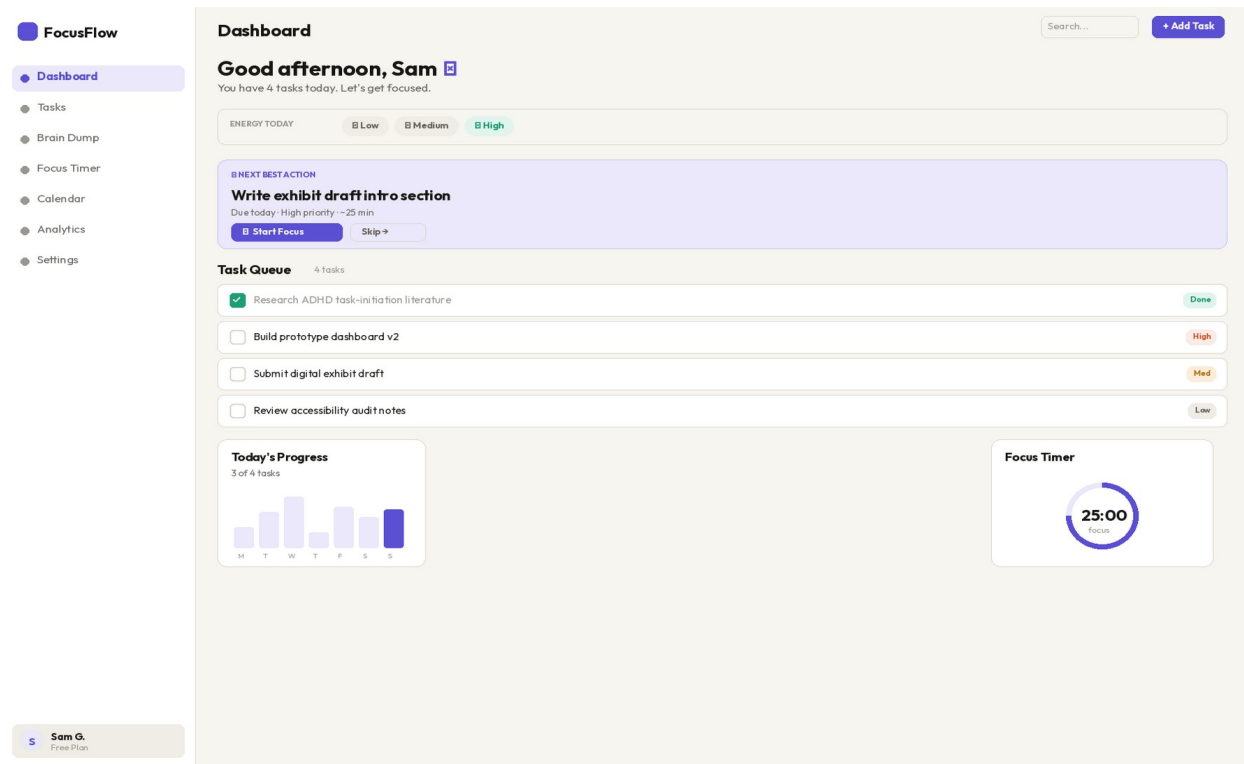


Fig. 1 — FocusFlow main dashboard. The interface surfaces a single recommended task based on the user's current energy level.

EXHIBIT OVERVIEW

A Cognitive Operating System for the ADHD Brain

SHORT EXHIBIT LABEL

FocusFlow is a web-based augmentation prototype I designed to reduce executive function friction for users with ADHD and related neurodivergent profiles. It does not ask users to manage their attention. It manages the cognitive scaffolding for them.

Most productivity tools are built on the assumption that users already know what they need to do, and simply need a place to write it down. For people with attention deficit hyperactivity disorder, that assumption collapses immediately. I know this because I have lived it. The challenge is rarely forgetting a task. It is initiating one, sustaining focus through it, and deciding which of several competing demands deserves attention right now.

I built FocusFlow on a different premise. Rather than presenting a blank canvas or an overwhelming list, it surfaces a single clear next action based on the user's current energy level, deadlines, and task

priority. It reduces what researchers call decision fatigue, one of the most significant cognitive barriers facing people with executive dysfunction, by removing the meta-work of figuring out where to start.

WHAT IT AUGMENTS

Executive Function, Not Just Productivity

Executive function is an umbrella term for the cognitive processes that regulate goal-directed behavior. This includes working memory, cognitive flexibility, inhibitory control, task initiation, and planning. These are not soft skills or habits. They are neurological processes that operate differently in ADHD brains, regardless of intelligence, motivation, or effort.

FocusFlow augments the specific moments where executive function tends to break down: the gap between knowing something needs to be done and actually starting it. This is sometimes called the intention-action gap, and for many ADHD users, including myself, it represents hours of paralysis disguised as procrastination. I wanted to build something that intervened at that exact point.

Task Initiation Surfaces one recommended action based on current energy state, cutting through the full task list to eliminate initiation paralysis.	Decision Fatigue Reducing the choices presented at once lowers the cognitive overhead of beginning any given work session.
Working Memory The Brain Dump module externalizes thoughts instantly, freeing working memory from holding half-formed ideas.	Time Blindness Integrated focus timers and a visual deadline calendar address the ADHD experience of time as either now or not now.

WHO IT'S FOR

Centered on Neurodivergent Users, Useful to Many

The primary audience I had in mind when building FocusFlow is adults with ADHD, particularly college students and knowledge workers navigating environments that reward sustained attention and consistent output. These users often have high intelligence and genuine motivation, yet find themselves unable to translate intention into action in the ways productivity culture assumes they should.

I also designed for users with executive dysfunction that is not formally diagnosed. Many people experience task-initiation difficulty, overwhelm from complex lists, or inconsistent focus without

ever receiving an ADHD diagnosis. FocusFlow does not require a label to be useful. I built it to reduce friction for anyone whose brain does not work on a linear schedule.

University disability services offices, ADHD coaches, and corporate employee assistance programs represent secondary audiences who might adopt FocusFlow as a digital accommodation tool. I see this as structured cognitive support at scale, without requiring one-on-one intervention for every session.

HOW IT WORKS

Adaptive Scaffolding Through a Unified Dashboard

I built FocusFlow as a single-file web application in HTML, CSS, and JavaScript. There is no backend required, no account creation, and no installation. A user opens the file and is immediately inside a working environment. This friction-free entry point is itself a design decision rooted in ADHD-aware thinking: every additional step between intention and use is a potential abandonment point. I wanted the barrier to entry to be as close to zero as possible.

Next Best Action Surfaces a single recommended task based on energy level and priority. I designed this to eliminate initiation paralysis by removing choice from the starting point.	Energy Check-In Users indicate current cognitive state before the session. Low, medium, and high energy modes adjust task recommendations. The tool adapts to the user.
Brain Dump A frictionless capture zone where any thought can be logged immediately without organizing first. I added this because categorizing before recording felt like its own barrier.	Focus Timer Configurable Pomodoro sessions with visual and ambient audio cues. Structures time into bounded intervals that I find easier to commit to than open-ended blocks.
AI Task Decomposition Vague tasks can be submitted to an AI assistant that breaks them into specific sub-steps. This addresses the experience of knowing a task exists without knowing how to begin it.	Deadline Calendar A visual timeline maps upcoming deadlines by proximity and priority. This counteracts time blindness by making future obligations spatially concrete rather than abstract.

Brain Dump

Frictionless thought capture without the pressure of organizing first.

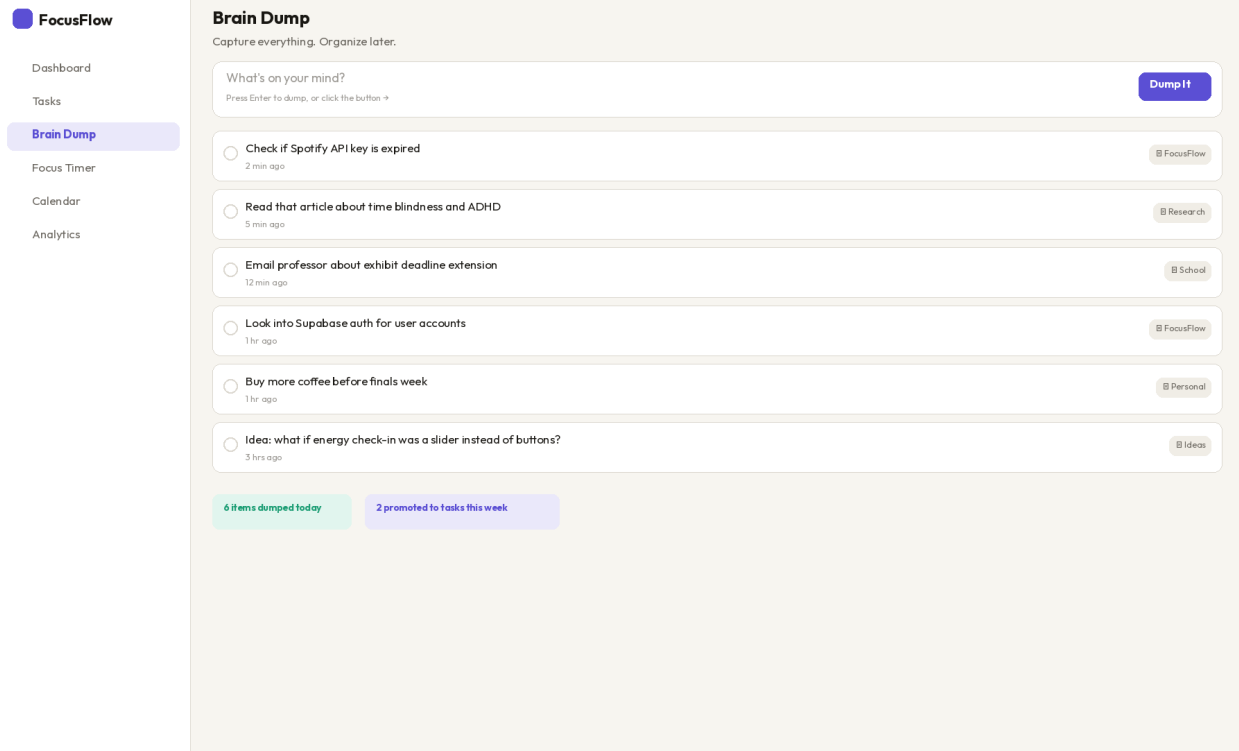


Fig. 2 — The Brain Dump module. Thoughts are captured immediately and tagged loosely. Organization happens later, or not at all.

PROCESS & CRITICAL MAKING

Built Through Iteration and Lived Frustration

FocusFlow did not begin as a product concept. It began as a personal frustration with every existing productivity tool on the market. I built the prototype out of direct experience of the problem it addresses. That starting point mattered, because it meant I evaluated design decisions not against what looked good in a feature list, but against what actually worked in a moment of executive dysfunction.

01 Problem Identification

My initial insight was that ADHD productivity tools almost universally imported frameworks from neurotypical productivity culture: full task lists, priority rankings, weekly planners. These formats presuppose that the act of choosing where to start is easy. For ADHD users, it is often the hardest part. I wanted to build something that started from that recognition.

02 First Prototype

My early version prioritized radical simplicity, showing a single task and a single timer. Testing revealed that users still struggled to load the task list in the first place, indicating that input friction was as much a barrier as output complexity. The Brain Dump feature came directly from that observation.

03 Energy-Adaptive Logic

The energy check-in system emerged from recognizing that my own ADHD symptoms fluctuate significantly across a single day. A task that is manageable at peak capacity may be impossible at low energy. Designing a tool that ignored that variance was designing for the exception, not the rule.

04 Visual Refinement

I went through multiple rounds of visual iteration focused on information density and calm. Dense UIs increase cognitive load. The final aesthetic, warm neutrals with restrained accent color, prioritizes focus rather than stimulation. I deliberately designed the tool not to be exciting to look at.

05 AI Integration

Adding AI-assisted task decomposition addressed a gap no visual design could close: some tasks are genuinely unclear in their first steps. Rather than expecting users to solve this themselves, the prototype offloads that cognitive work to a language model on request. I found this made a meaningful difference in my own usage.

Focus Timer

Bounded time intervals that make sustained attention feel achievable.

FocusFlow

Dashboard

Tasks

Brain Dump

Focus Timer

Calendar

Analytics

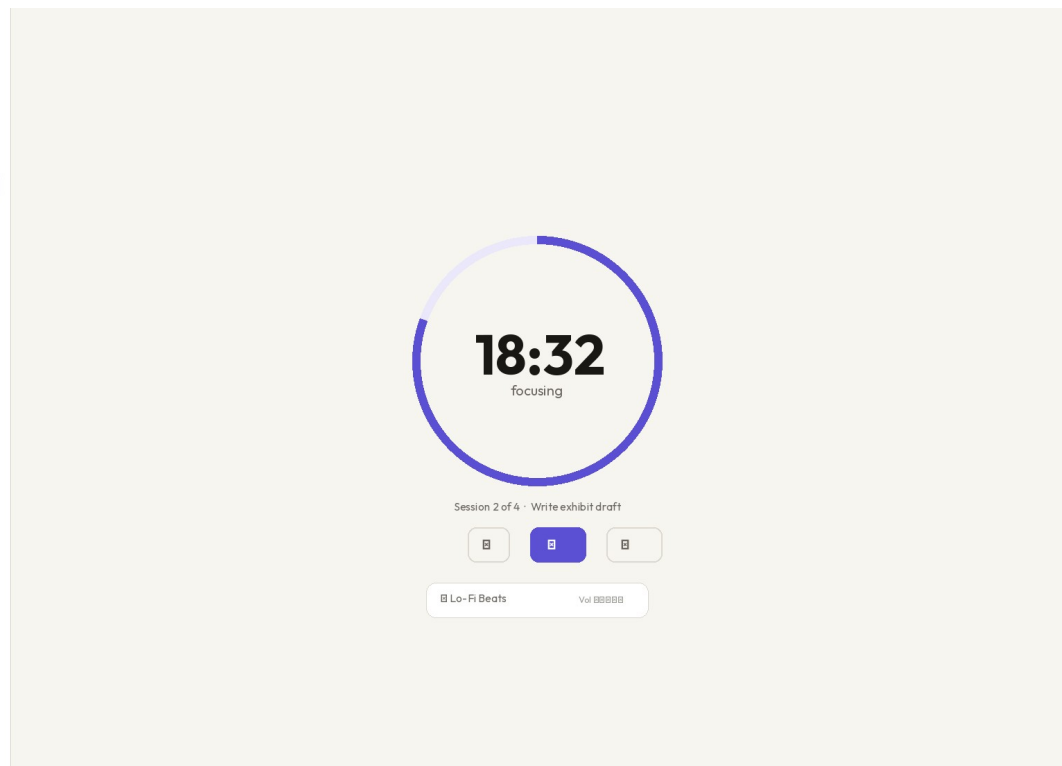


Fig. 3 — The Focus Timer mid-session. Configurable Pomodoro intervals with ambient audio integration and session tracking.

ACCESSIBILITY, USABILITY & USER EXPERIENCE

Designed Around How ADHD Brains Actually Work

Accessibility in FocusFlow is not a checklist I completed after the design was finished. It is the design. The distinction matters because conventional accessibility thinking focuses heavily on sensory and motor access, which is critical but incomplete for a neurodivergent audience.

Cognitive accessibility, to me, means reducing the number of decisions required to use the tool. It means making the interface readable under distraction. It means ensuring that a user who abandoned the app mid-session can return to it without losing context or having to reconstruct where they were. I built FocusFlow to hold state gracefully so that interrupted attention is not punished.

CORE DESIGN PRINCIPLE

I evaluated every feature against a single question: does this reduce cognitive load, or does it add to it? Features that scored well on a conventional usability rubric but increased mental overhead were cut or restructured. The interface defaults to the minimum viable information for any given context.

WHAT WORKS

The energy check-in removes the burden of self-calibration from task selection. Dark mode reduces visual strain during late-night sessions. Single-task focus view eliminates distraction from the full task queue. Keyboard accessibility is maintained throughout.

LIMITATIONS & TRADEOFFS

The prototype currently lacks screen reader optimization beyond semantic HTML. Mobile responsiveness exists but is not the primary design context. AI decomposition requires an active internet connection, creating a dependency.

HONEST LIMITATION

FocusFlow is a digital-only tool, which means it is inaccessible to users who struggle with sustained screen engagement or who benefit from physical, tactile interaction with their planning system. This is a real exclusion I am aware of. A future hybrid version pairing the digital interface with a printed daily card generated from the dashboard would address this gap directly.

WHY IT MATTERS

A Sociotechnical Argument About Whose Brains Get Designed For

Productivity software is rarely described as political. But every design choice encodes assumptions about what counts as a normal user. The calendar that starts the week on Monday assumes a person

who experiences time as consistent. The to-do list that expects hierarchical organization assumes a working memory that can hold structure. The focus timer that runs until you stop it assumes attention that can be applied by will.

These are not neutral design choices. They reflect a model of the mind that excludes a significant portion of the population. Approximately 5 to 8 percent of adults have diagnosable ADHD. A much larger number experience executive dysfunction without a formal diagnosis. When the dominant tools of knowledge work are designed exclusively for neurotypical cognitive profiles, the result is not just inconvenience. It is a systemic barrier to participation.

I built FocusFlow as an argument, in the form of a prototype, that design can meet users where they are rather than demanding they adapt to the tool. It applies the values of user-centered design to an audience that is frequently told their difficulty with standard tools is a personal failure rather than a design failure.

It also raises questions that my prototype does not fully answer. Augmentation technologies can support users without addressing the structural conditions that create friction in the first place. A better task manager does not change the fact that many workplaces and academic environments are built on an attention economy that disadvantages neurodivergent people. FocusFlow is a useful tool. It is not a solution to a sociotechnical problem that runs deeper than any single interface can reach.

"The question is not whether ADHD brains can perform. They often perform extraordinarily well." The question I wanted FocusFlow to ask is why the tools designed to support performance consistently assume a brain that does not need support in the first place. Augmentation technologies are most meaningful when they refuse to treat difference as deficiency.

TRY IT OUT

Try FocusFlow for yourself by scanning the QR code below or clicking this link:
<https://app.netlify.com/projects/teal-boba-954873/overview>.

